

1

Governed Bilingual Inputs



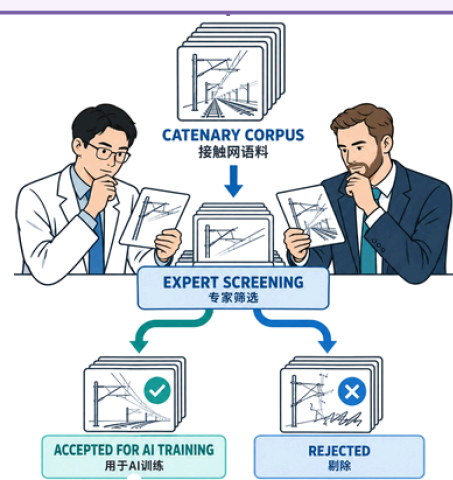
q_{zh}: 接触网供电的额定电压是多少

parallel inputs

q_{en}: What is the rated voltage for contact network power supply

Governed bilingual records

Chinese/English · source review · split



Admissibility filter

approved / test excluded

Query-time approval only; not a trainable mask.

2

Shared Neural Encoding

Frozen shared BGE-M3 encoder

shared query/evidence parameters

Token + positional embeddings

Multi-Head Self-Attention

Q

K

V

Linear

Linear

Linear

Scaled Dot-Product Attention

Concat

Linear

Residual + Add & Norm

Feed-Forward Network

Residual + Add & Norm x N

$z_q = \text{Encoder}_{\text{BGE-M3}}(q)$

$z_d = \text{Encoder}_{\text{BGE-M3}}(d)$

$z_q, z_d \in \mathbb{R}^{1024}$

3

Parallel Hybrid Retrieval

Embedding Matrix

0.11

0.86

0.65

0.64

0.33

0.19

Dense retrieval

$\cos(z_q, z_d)$

vector rank

BM25 retrieval

lexical rank

Reciprocal Rank Fusion

$\text{RRF}(d) = \sum 1/(k_0 + \text{rank})$

Top-k evidence passages

[E1]/[E2] passage + source

4

Evidence-Conditioned LLM + QLoRA

Serialized context

[Instr] | [Q] | evidence | [A]

Repeated decoder

Q

K

V

Masked causal attention

Residual + Add & Norm

Gated FFN

Residual + Norm x N

state

QLoRA

low-rank update

x

W_0

A

$r \times d$

B

$d \times r$

$+$

$dW = BA$

$r = 64; \alpha = 16$

One shared LM head + Softmax

$p(y_t | X, y_{<t})$

Autoregressive bilingual answer

token1 -> token2 -> token3 -> ...

evidence labels: [E1] / [证据1]

Independent evaluation

external metrics / no gradient

Railway QA

F1 / containment

Citations

coverage / syntax

Translation

BLEU / chrF / COMET

Evidence support

semantic proxy

Resource-constrained inference

latency / GPU memory / tokens/s